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CO2 AT1 SCENARIO BASED

AUTONOMOUS ELECTRIC VEHICLE CHARGING NETWORK MANAGEMENT SYSTEM

1. Introduction

Purpose

This document presents the Software Requirement Specification (SRS) for the Autonomous Electric Vehicle Charging Network Management System (AEVCNMS). It defines the functional and non-functional requirements, actors, use cases, data needs, and constraints for a platform that manages a distributed network of smart, self-monitoring EV charging stations and supports autonomous vehicles that can locate, navigate to, and dock at charging points with minimal human intervention.

The system operates in the smart mobility and clean energy infrastructure domain, at the intersection of electric vehicle transportation, IoT-enabled charging hardware, energy grid management, and autonomous navigation technology.

Intended Audience

This document is intended for the development team, project stakeholders, quality assurance engineers, charging network operators, utility partners, and academic evaluators reviewing the requirement analysis.

Definitions, Acronyms and Abbreviations

AEVCNMS	Autonomous Electric Vehicle Charging Network Management System
AV	Autonomous Vehicle capable of self-navigation to a charging point
DERMS	Distributed Energy Resource Management System (utility-side platform)

SoC	State of Charge – remaining battery capacity of an EV
OCPP	Open Charge Point Protocol – industry standard for charger communication

2. Problem Description

Electric vehicle adoption is growing rapidly, but existing charging infrastructure is fragmented: station availability is not visible in real time, pricing is static regardless of grid load, reservations are unreliable, and there is no coordinated mechanism to guide autonomous or driver-assisted vehicles to an open, functioning charger. Station operators lack centralized visibility into faults, utilization, and energy demand, which leads to downtime, uneven load on the local grid, and poor user experience. There is a need for an intelligent, network-wide management system that can autonomously monitor station health, guide vehicles (including self-driving vehicles) to available chargers, balance load across the network, apply dynamic pricing, and coordinate with the utility grid — all while giving drivers and administrators a reliable, real-time view of the network.

Existing Challenges

- No unified real-time visibility into charger availability, health, or utilization across a network.
- Manual, reactive fault detection leads to prolonged station downtime.
- Static, one-size pricing does not reflect grid load, time-of-day demand, or renewable energy availability.
- Autonomous and driver-assisted vehicles lack a standard mechanism to be guided to and precisely docked at an available charger.
- No coordinated load balancing, causing localized grid strain during peak charging periods.
- Fragmented payment and reservation experiences across different station vendors.

System Objectives

- Provide real-time discovery, reservation, and status tracking of charging stations.
- Support autonomous vehicle navigation, docking, and session handling with minimal human input.
- Apply dynamic, demand-based pricing and coordinate with utility demand-response signals.
- Continuously monitor station health, detect faults automatically, and trigger maintenance workflows.

- Balance charging load across the network to protect grid stability and maximize station uptime.
- Provide administrators and utility partners with analytics on usage, revenue, and grid impact.

3. Scope of the System

The AEVCNMS will cover station discovery and reservation, session authentication and metering, autonomous vehicle guidance and docking assistance, dynamic pricing, network-wide load balancing, automated fault detection and maintenance dispatch, payment processing, and integration with utility grid platforms. The system includes a driver-facing mobile/web application, an administrator dashboard, an autonomous-vehicle integration API, and backend services for pricing, load management, and analytics.

Out of scope: manufacturing or physical installation of charging hardware, vehicle battery management internals, and city-wide traffic routing outside the immediate vicinity of charging stations.

4. Stakeholders

- **EV Users** – Use the system to find and book charging stations.
- **Charging Station Operators** – Manage and maintain charging stations.
- **System Administrators** – Monitor and control the system operations.
- **Payment Gateway Providers** – Handle secure online payments.
- **Energy Providers** – Supply electricity and manage energy usage.
- **Government Authorities** – Ensure legal and regulatory compliance.
- **Maintenance Team** – Handle repairs and ensure proper functioning of stations.

5. Functional Requirements

1. The system shall allow registered users to search for and view nearby charging stations with real-time availability, connector type, and pricing.
2. The system shall allow users to reserve a charging slot at a selected station for a specified time window.
3. The system shall authenticate users (via app, RFID, or vehicle credentials) before starting a charging session.
4. The system shall provide an API through which autonomous vehicles can request the nearest available compatible station and receive navigation/docking coordinates.

5. The system shall meter energy delivered in real time and display live session status (SoC, time remaining, cost accrued) to the user.

Data Requirements

- User Account Data: name, contact details, payment methods, vehicle profile(s), authentication credentials.
- Vehicle Data: vehicle ID, connector type, current SoC, autonomous-navigation capability flag.
- Station Data: station ID, location (GPS), connector types, real-time status, power rating, telemetry (temperature, fault codes).
- Session Data: session ID, start/end time, energy delivered, cost, payment status, user and station references.

9. External Interface Requirements

- Mobile application and responsive web portal for drivers (station search, reservation, live session view, payment).
- Administrator web dashboard (network map, station management, alerts, analytics).
- Charging station controllers communicating via OCPP over a secure network connection.
- Sensors for connector state, temperature, and fault detection at each station.
- Vehicle telematics / autonomous-navigation API for AV integration.
- Payment gateway API for transaction authorization and settlement.
- Utility DERMS API for demand-response signal exchange and load reporting.
- REST/JSON APIs secured with TLS and OAuth 2.0 token-based authentication.
- Push notification, SMS, and email gateways for user alerts.

10. System Features

- **User Registration & Login** – Allows users to create accounts and securely access the system.
- **Charging Station Locator** – Displays nearby charging stations using GPS.
- **Real-Time Availability** – Shows available charging slots instantly.
- **Slot Booking System** – Enables users to reserve charging slots in advance.

- **Live Charging Monitoring** – Tracks charging status in real time.
- **Payment Integration** – Supports online payments (UPI, card, wallet).
- **Notifications & Alerts** – Sends updates for booking, charging status, and delays.
- **Admin Dashboard** – Allows administrators to manage stations, users, and reports.
- **Energy Usage Tracking** – Monitors electricity consumption at each station.

11. Assumptions and Constraints

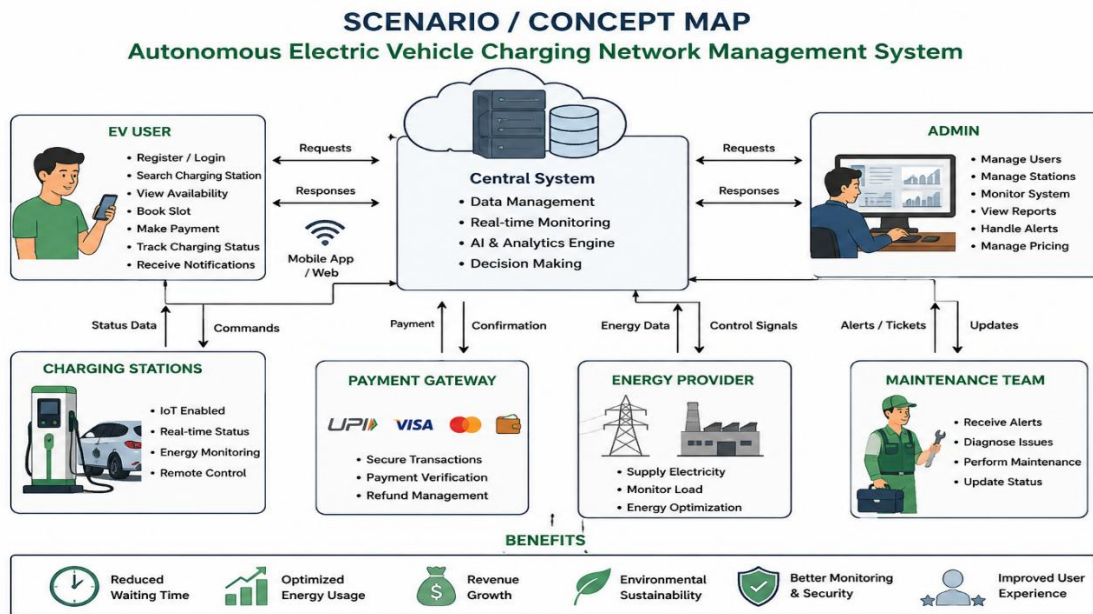
- Users have internet-enabled smartphones or in-vehicle systems capable of running the client application.
- Participating vehicles are equipped with compatible connectors and, for autonomous docking, compatible navigation/telematics hardware.
- Charging stations have reliable network connectivity to report telemetry in real time.
- The utility/grid operator's DERMS platform exposes a documented API for demand-response integration.
- Payment gateway services used are PCI-DSS compliant and available via API.

12. Requirement Completeness and Consistency Validation

Each stakeholder group identified in Section 4 maps to at least one functional requirement and use case: drivers (UC-01, UC-02, UC-03, UC-05), autonomous vehicles (UC-04), administrators (UC-09, UC-11, UC-12), maintenance technicians (UC-09), the payment gateway (UC-07), and the utility operator (UC-10). No stakeholder identified in the analysis is left without a corresponding requirement.

- **Ambiguity:** The term “near-real-time” was replaced with explicit thresholds (2-second query response, 5-second telemetry propagation) to remove vagueness.
- **Redundancy:** Station monitoring and fault detection were consolidated into a single non-functional/functional pairing (FR-9 and FR-10) rather than duplicated across sections.
- **Inconsistency:** An initial draft implied both the vehicle and the station independently confirm session start; this was resolved so the station's physical-connection confirmation is the single source of truth (see UC-04 postcondition).

- Missing Requirements Identified: Explicit handling for reservation timeout/no-show and automatic session rollback on payment failure were added after review (FR-2 alternate flow, Reliability NFR).



I V. PAVITHRA (19242524274) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action

V. Pavithra